Nonlinear Model Predictive Control of a Four-Node District Heating Network

Prepared by: Muhammad Kashif Sarwar
Date: March 10, 2026

Abstract

This report presents the modelling, nonlinear model predictive control formulation, simulation setup, and result interpretation for a simplified four-node district heating network (DHN). The network is composed of a producer node, two pipe sections, and a consumer node arranged in a closed thermal ring. A nonlinear model predictive controller is designed in MATLAB/Simulink using a continuous-time plant model and custom cost and constraint functions. The manipulated variables are the four circulation flow rates and the producer heat input, while the consumer heat demand acts as a measured disturbance. The report first introduces the physical meaning of the system and then derives the governing differential equations from a general thermal energy balance. It subsequently formulates the NMPC problem with its cost function, equality constraints, and inequality constraints, and explains the closed-loop simulation implementation. The reported results show successful optimization convergence and near-perfect power matching between the producer input and the time-varying heat demand profile. At the same time, the temperature trajectories reveal that the present controller formulation does not explicitly regulate the node temperatures to the provided reference values, which leads to an important discussion about the difference between demand-following and temperature-tracking control objectives.

Contents

1 Introduction **1**

1.1 Background and motivation 1

1.2 System description 1

1.3 Objectives of the study 3

1.4 Report organization 3

2 System Modelling **4**

2.1 General modelling philosophy 4

2.2 Node-wise differential equations 5

2.2.1 Node 1: producer node 6

2.2.2 Node 2: supply pipe section 6

2.2.3 Node 3: consumer node 6

2.2.4 Node 4: return pipe section 6

2.3 Matrix form of the model 6

2.4 Interpretation of the derivative terms 10

2.5 Physical parameterization used in the code 11

3 NMPC Formulation **14**

3.1 Controller structure 14

3.2 State and output functions 15

3.3 Objective function 15

3.4 Equality constraints 16

3.5 Inequality constraints 16

3.6 Complete NMPC problem statement 17

3.7 Discussion of the chosen formulation 18

4 Simulation Setup **19**

4.1 Closed-loop implementation in MATLAB/Simulink 19

4.2 Measured disturbance profile 19

4.3 Plant subsystem 20

4.4 Initial conditions and references 20

4.5 Solver configuration 21

4.6 Expectation before simulation 21

5 Results and Discussion **22**

5.1 Overview of obtained closed-loop behavior 22

5.2 Optimization status 22

5.3 Flow-rate trajectories 23

5.4 Power and disturbance trajectories 26

5.5 Temperature trajectories 27

5.6 Discussion 29

5.6.1 Observation 1: the NMPC implementation is numerically successful 30

5.6.2 Observation 2: power tracking works exactly as designed 30

5.6.3 Observation 3: equality constraints dominate the flow behavior . . 30

5.6.4 Observation 4: temperature references are not being controlled . . 30

5.6.5 Interpretation 30

6 Conclusion **32**

Bibliography **34**

List of Figures

1.1 Overall Simulink implementation of the four-node DHN with nonlinear MPC controller, measured disturbance generation, and plant subsystem. 2

2.1 Schematic representation of the four-node district heating network model showing the producer, supply pipe, consumer, and return pipe nodes. . . 5

4.1 Plant subsystem used in Simulink. The MATLAB function block computes the temperature derivatives, which are then integrated to produce the state vector. 20

5.1 Optimization status during the full 24-hour simulation. 23

5.2 Manipulated flow rate q_{12} from node 1 to node 2. 24

5.3 Manipulated flow rate q_{23} from node 2 to node 3. 24

5.4 Manipulated flow rate q_{34} from node 3 to node 4. 25

5.5 Manipulated flow rate q_{41} from node 4 to node 1. 25

5.6 Producer heat input P over the 24-hour simulation. 26

5.7 Measured disturbance profile P_d applied at the consumer node. 26

5.8 Temperature trajectory at node 1, T_1. 27

5.9 Temperature trajectory at node 2, T_2. 28

5.10 Temperature trajectory at node 3, T_3. 28

5.11 Temperature trajectory at node 4, T_4. 29

List of Tables

2.1 State, manipulated-variable, and disturbance definitions used in the DHN model. 11

2.2 Physical properties and node volumes used in the DHN model. 12

2.3 Heat-loss coefficients used in the DHN model. 13

3.1 Key NMPC tuning settings used in the simulation. 15

Chapter 1

Introduction

1.1 Background and motivation

District heating networks transport thermal energy from one or more heat producers to distributed consumers through a hydraulic and thermal circulation loop. In practical systems, the interaction between water transport, thermal inertia, heat demand, and environmental heat loss creates a dynamic system that is naturally multivariable and constrained. For this reason, district heating networks are well suited for predictive control methods, especially model predictive control (MPC), because such methods can explicitly account for future disturbances, plant dynamics, and physical operating limits.

In the present work, a reduced-order but physically interpretable district heating network is considered. The objective is not to build a full-scale urban DHN model with many pipelines and distributed substations, but rather to construct a compact closed-loop benchmark that captures the main mechanisms needed for nonlinear model predictive control design. The model has four thermal states and is implemented in MATLAB/Simulink using a continuous-time plant representation inside a nonlinear MPC framework.

1.2 System description

The considered district heating network consists of four thermal nodes arranged in a ring:

- Node 1: producer side heat exchanger, where external heat input is injected into the water stream,
- Node 2: upstream pipe section,
- Node 3: consumer side thermal element, where heat demand is extracted,
- Node 4: downstream pipe section returning the fluid back to the producer.

The thermal state vector is defined as

$$x(t) = \begin{bmatrix} T_1(t) & T_2(t) & T_3(t) & T_4(t) \end{bmatrix}^\top, \tag{1.1}$$

where each state represents the average water temperature at one node. In the MATLAB implementation, the initial states and reference temperatures is chosen as

$$x(0) = \begin{bmatrix} 85 & 85 & 65 & 65 \end{bmatrix}^\top \ ^\circ\mathrm{C}. \tag{1.2}$$

The manipulated variables of the controller are

$$u(t) = \begin{bmatrix} q_{12}(t) & q_{23}(t) & q_{34}(t) & q_{41}(t) & P(t) \end{bmatrix}^\top, \tag{1.3}$$

where q_{ij} denotes the mass flow rate from node i to node j and P denotes the producer heat input. The measured disturbance is the consumer heat demand $P_d(t)$, which enters at node 3. In the controller implementation, the total input vector passed to the plant model is

$$\tilde{u}(t) = \begin{bmatrix} q_{12}(t) & q_{23}(t) & q_{34}(t) & q_{41}(t) & P(t) & P_d(t) \end{bmatrix}^\top. \tag{1.4}$$

The structure of the Simulink implementation is shown in Figure 1.1. The nonlinear MPC block receives the current temperatures, the reference vector, the last applied input, and the measured disturbance. Its outputs are the manipulated variables and the optimization status. The resulting flow rates and producer heat input are applied to the plant subsystem, while the measured disturbance is generated from a separate time-varying function block.

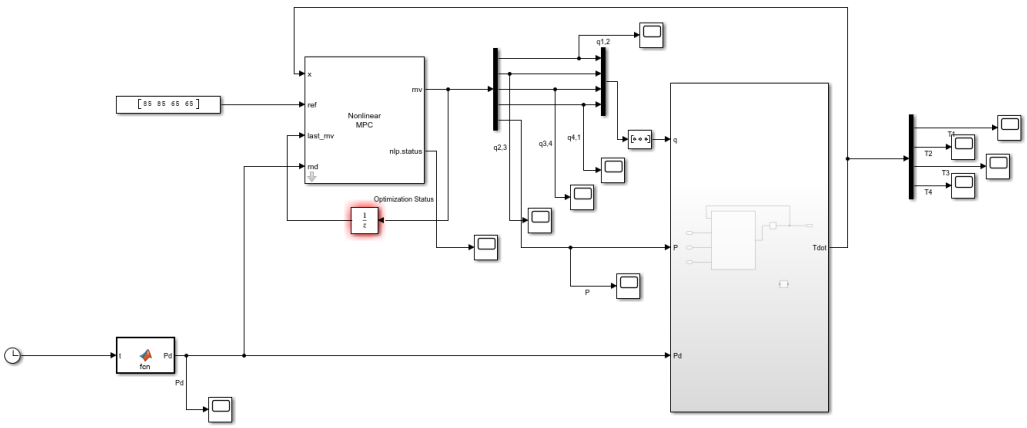

Figure 1.1: Overall Simulink implementation of the four-node DHN with nonlinear MPC controller, measured disturbance generation, and plant subsystem.

1.3 Objectives of the study

The main objectives of the study can be stated as follows:

1. to formulate a compact dynamic model of a four-node district heating network using physically meaningful thermal balances,
2. to embed this model inside a nonlinear model predictive control framework in MATLAB/Simulink,
3. to define a custom optimization problem using a user-defined cost function and explicit equality and inequality constraints,
4. to simulate the closed-loop response under a realistic daily heat-demand profile,
5. to evaluate the resulting control trajectories and thermal state evolution, and
6. to discuss what the obtained results imply about the present NMPC design and how it may be improved.

1.4 Report organization

Chapter 1 introduces the system and its physical meaning. Chapter 2 develops the system model from the general energy-balance principle and presents the detailed differential equations used in the code. Chapter 3 formulates the nonlinear MPC problem, including the cost function and constraints. Chapter 4 explains the simulation setup and the Simulink realization. Chapter 5 presents the obtained results and discusses their physical and control-theoretic interpretation. Finally, Chapter 6 summarizes the main findings and concludes the report.

Chapter 2

System Modelling

2.1 General modelling philosophy

The four-node DHN model is built from a lumped-parameter thermal description. Each node is treated as a well-mixed control volume with a single representative temperature. This modelling assumption is common in reduced-order thermal systems, because it preserves the dominant energy-storage dynamics while keeping the state dimension low enough for online predictive control.

Starting from the energy balance written in enthalpy form for node i,

$$m_i c_p \dot{T}_i = \sum \dot{m}_{in} c_p T_{in} - \sum \dot{m}_{out} c_p T_i - UA_i(T_i - T_{a,i}) + Q_i, \tag{2.1}$$

where Q_i denotes the external heat injection or extraction at node i.

Dividing Equation (2.1) by the specific heat capacity c_p yields

$$m_i \dot{T}_i = \sum \dot{m}_{in} T_{in} - \sum \dot{m}_{out} T_i - \frac{UA_i}{c_p}(T_i - T_{a,i}) + \frac{Q_i}{c_p}. \tag{2.2}$$

To simplify notation, the auxiliary heat-loss coefficient is introduced as

$$\kappa_i := \frac{UA_i}{c_p}. \tag{2.3}$$

Substituting this definition into Equation (2.2) gives the temperature-coordinate model

$$m_i \dot{T}_i = \sum \dot{m}_{in} T_{in} - \sum \dot{m}_{out} T_i - \kappa_i(T_i - T_{a,i}) + \frac{Q_i}{c_p}. \tag{2.4}$$

It is important to note that the factor $1/c_p$ appears explicitly only in the external heat-

input term Q_i. The convective mixing terms originally contain the factor c_p, which cancels when the equation is divided by c_p. Similarly, the heat-loss term is rewritten using the coefficient $\kappa_i = UA_i/c_p$, which absorbs the factor $1/c_p$ into the loss parameter.

The code defines the auxiliary coefficient

$$\kappa_i = \frac{UA_i}{c_p}, \tag{2.5}$$

so that the governing equations may be written directly in temperature coordinates. This is exactly the structure used in the MATLAB function `DHN_ct_plant`.

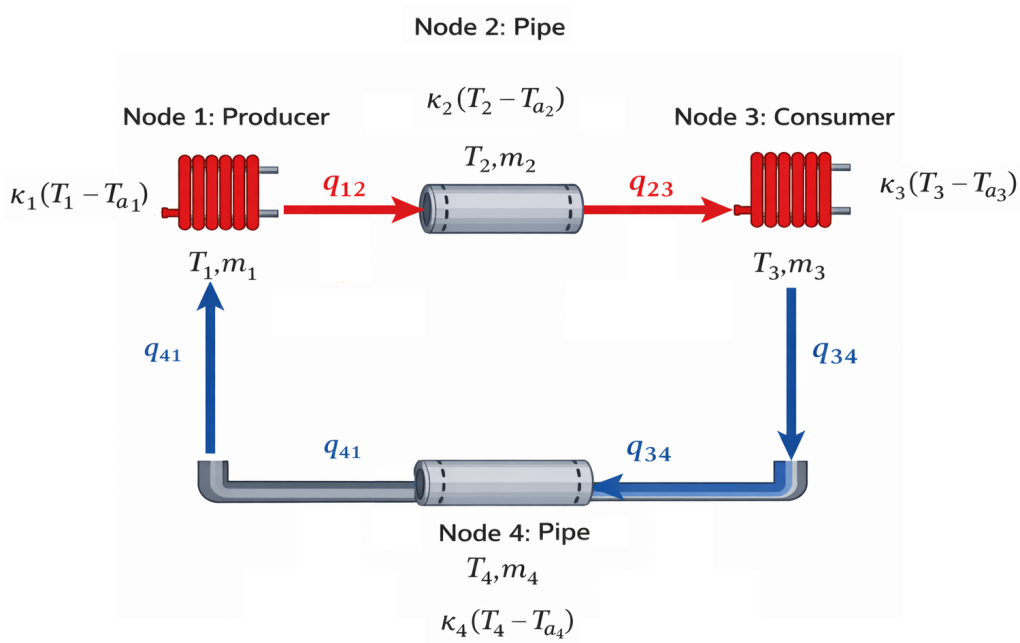

Figure 2.1: Schematic representation of the four-node district heating network model showing the producer, supply pipe, consumer, and return pipe nodes.

2.2 Node-wise differential equations

The network topology is a ring with the directed flow sequence

$$1 \to 2 \to 3 \to 4 \to 1. \tag{2.6}$$

Accordingly, the node equations become:

2.2.1 Node 1: producer node

At node 1, fluid enters from node 4 through flow q_{41} and exits toward node 2 through flow q_{12}. In addition, thermal power P is injected at this node. Therefore,

$$m_1\dot{T}_1 = q_{41}T_4 - q_{12}T_1 - \kappa_1 T_1 + \kappa_1 T_{a,1} + \frac{P}{c_p}. \tag{2.7}$$

2.2.2 Node 2: supply pipe section

Node 2 receives heat through the stream from node 1 and loses heat to the surroundings. Its equation is

$$m_2\dot{T}_2 = q_{12}T_1 - q_{23}T_2 - \kappa_2 T_2 + \kappa_2 T_{a,2}. \tag{2.8}$$

2.2.3 Node 3: consumer node

At node 3, thermal power is extracted in the form of demand P_d. The equation is

$$m_3\dot{T}_3 = q_{23}T_2 - q_{34}T_3 - \kappa_3 T_3 + \kappa_3 T_{a,3} - \frac{P_d}{c_p}. \tag{2.9}$$

The negative sign in front of P_d/c_p reflects that demand removes heat from the network rather than injecting it.

2.2.4 Node 4: return pipe section

Finally, the return pipe section is modelled as

$$m_4\dot{T}_4 = q_{34}T_3 - q_{41}T_4 - \kappa_4 T_4 + \kappa_4 T_{a,4}. \tag{2.10}$$

2.3 Matrix form of the model

The four-node district heating network model is obtained by writing the temperature dynamics of each node individually and then collecting them into a compact state-space form.

Using the lumped thermal balance for each node, the governing equations are

$$m_1\dot{T}_1 = q_{41}T_4 - q_{12}T_1 - \kappa_1(T_1 - T_{a,1}) + \frac{P}{c_p}, \tag{2.11}$$

$$m_2\dot{T}_2 = q_{12}T_1 - q_{23}T_2 - \kappa_2(T_2 - T_{a,2}), \tag{2.12}$$

$$m_3\dot{T}_3 = q_{23}T_2 - q_{34}T_3 - \kappa_3(T_3 - T_{a,3}) - \frac{P_d}{c_p}, \tag{2.13}$$

$$m_4\dot{T}_4 = q_{34}T_3 - q_{41}T_4 - \kappa_4(T_4 - T_{a,4}). \tag{2.14}$$

Each equation contains four physically meaningful contributions:

- the inflowing thermal transport term,
- the outflowing thermal transport term,
- the heat loss to the ambient environment,
- and, where applicable, an external heat input or heat extraction term.

To make this structure clearer, each equation may first be expanded by distributing the heat-loss term:

$$m_1\dot{T}_1 = q_{41}T_4 - q_{12}T_1 - \kappa_1 T_1 + \kappa_1 T_{a,1} + \frac{P}{c_p}, \tag{2.15}$$

$$m_2\dot{T}_2 = q_{12}T_1 - q_{23}T_2 - \kappa_2 T_2 + \kappa_2 T_{a,2}, \tag{2.16}$$

$$m_3\dot{T}_3 = q_{23}T_2 - q_{34}T_3 - \kappa_3 T_3 + \kappa_3 T_{a,3} - \frac{P_d}{c_p}, \tag{2.17}$$

$$m_4\dot{T}_4 = q_{34}T_3 - q_{41}T_4 - \kappa_4 T_4 + \kappa_4 T_{a,4}. \tag{2.18}$$

Now the state vector is introduced as

$$x(t) = \begin{bmatrix} T_1(t) \\ T_2(t) \\ T_3(t) \\ T_4(t) \end{bmatrix}. \tag{2.19}$$

Similarly, the thermal mass matrix is defined as the diagonal matrix

$$M = \mathrm{diag}(m_1, m_2, m_3, m_4) = \begin{bmatrix} m_1 & 0 & 0 & 0 \\ 0 & m_2 & 0 & 0 \\ 0 & 0 & m_3 & 0 \\ 0 & 0 & 0 & m_4 \end{bmatrix}. \tag{2.20}$$

With this notation, the left-hand side of the four scalar equations can be written compactly as

$$M\dot{x}(t) = \begin{bmatrix} m_1\dot{T}_1 \\ m_2\dot{T}_2 \\ m_3\dot{T}_3 \\ m_4\dot{T}_4 \end{bmatrix}. \tag{2.21}$$

Next, the terms multiplying the state temperatures T_1, T_2, T_3, T_4 are grouped into a single matrix. From Equations (2.11)–(2.14), the coefficients of the state variables are

$$m_1\dot{T}_1: \quad (-q_{12} - \kappa_1)T_1 + 0 \cdot T_2 + 0 \cdot T_3 + q_{41}T_4, \tag{2.22}$$
$$m_2\dot{T}_2: \quad q_{12}T_1 + (-q_{23} - \kappa_2)T_2 + 0 \cdot T_3 + 0 \cdot T_4, \tag{2.23}$$
$$m_3\dot{T}_3: \quad 0 \cdot T_1 + q_{23}T_2 + (-q_{34} - \kappa_3)T_3 + 0 \cdot T_4, \tag{2.24}$$
$$m_4\dot{T}_4: \quad 0 \cdot T_1 + 0 \cdot T_2 + q_{34}T_3 + (-q_{41} - \kappa_4)T_4. \tag{2.25}$$

Therefore, the state-coupling matrix is

$$L(q) = \begin{bmatrix} -q_{12} - \kappa_1 & 0 & 0 & q_{41} \\ q_{12} & -q_{23} - \kappa_2 & 0 & 0 \\ 0 & q_{23} & -q_{34} - \kappa_3 & 0 \\ 0 & 0 & q_{34} & -q_{41} - \kappa_4 \end{bmatrix}. \tag{2.26}$$

Equivalently, this matrix may be written as the sum of the transport contribution and the ambient-loss contribution:

$$L(q) = \begin{bmatrix} -q_{12} & 0 & 0 & q_{41} \\ q_{12} & -q_{23} & 0 & 0 \\ 0 & q_{23} & -q_{34} & 0 \\ 0 & 0 & q_{34} & -q_{41} \end{bmatrix} - \mathrm{diag}(\kappa_1, \kappa_2, \kappa_3, \kappa_4). \tag{2.27}$$

This decomposition is useful because it separates two distinct physical mechanisms:

- the first matrix captures temperature transport caused by the circulating mass flow

rates,

- while the diagonal matrix accounts for local heat losses at each node.

The external heating power P enters only at node 1. Hence, its input distribution vector is

$$B = \frac{1}{c_p} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}. \tag{2.28}$$

Likewise, the thermal demand P_d is extracted only at node 3, so its distribution vector is

$$B_d = \frac{1}{c_p} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}. \tag{2.29}$$

The ambient-temperature contributions appear as additive offset terms in each scalar equation, namely

$$\begin{bmatrix} \kappa_1 T_{a,1} \\ \kappa_2 T_{a,2} \\ \kappa_3 T_{a,3} \\ \kappa_4 T_{a,4} \end{bmatrix}. \tag{2.30}$$

The external heating power injected at node 1 can be written as

$$BP(t) = \frac{1}{c_p} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} P(t), \tag{2.31}$$

where $P(t)$ denotes the thermal power supplied by the producer (e.g., a boiler or heat source). The vector B distributes this injected heat to the state equations, indicating that the control input acts only on the first node.

Combining the demand extraction term and the ambient-temperature offsets, the disturbance vector is defined as

$$d(P_d, T_a) = -B_d P_d + \begin{bmatrix} \kappa_1 T_{a,1} \\ \kappa_2 T_{a,2} \\ \kappa_3 T_{a,3} \\ \kappa_4 T_{a,4} \end{bmatrix}. \tag{2.32}$$

Substituting all of these components into the stacked system gives

$$M\dot{x}(t) = L(q)x(t) + BP(t) + d(P_d(t), T_a). \tag{2.33}$$

Finally, premultiplying by M^{-1} yields the compact nonlinear state model

$$\dot{x}(t) = M^{-1}\left(L(q)x(t) + BP(t) + d(P_d(t), T_a)\right). \tag{2.34}$$

The model is nonlinear because the matrix $L(q)$ depends on the mass flow rates $q_{12}, q_{23}, q_{34}, q_{41}$, which are treated as manipulated inputs in the predictive control formulation. Thus, although the state appears linearly inside the term $L(q)x$, the overall dynamics remain bilinear or nonlinear with respect to the combined state-input pair.

The output equation used by the NMPC block is simply

$$y(t) = x(t), \tag{2.35}$$

which means that all states are measured outputs.

2.4 Interpretation of the derivative terms

It is useful to interpret every term in the derivative expression because this makes the structure of the code physically transparent.

First, the terms of the form $q_{ij}T_j$ represent convective transport into a node. For example, the quantity $q_{41}T_4$ in Equation (2.7) says that cold water arriving from node 4 carries its thermal content into node 1. Second, the terms of the form $-q_{ij}T_i$ represent convective outflow from the node itself. Third, the terms $-\kappa_i T_i + \kappa_i T_{a,i}$ combine into $-\kappa_i(T_i - T_{a,i})$, which is a linearized heat loss to the environment. Finally, P/c_p and $-P_d/c_p$ represent source and sink terms due to injected or extracted thermal power.

Because the matrix $L(q)$ depends on the flow rates, the model is nonlinear with respect to the full input vector. Even though the state equations look linear in x for fixed q, the dependence of L on the manipulated flow variables makes the plant input-dependent and therefore nonlinear from the control point of view.

2.5 Physical parameterization used in the code

The parameter set is implemented in the MATLAB function `DHN_defaultParams`. Water is assigned a constant heat capacity of $c_p = 4180$ J/(kg K). Node densities are selected according to two nominal temperature levels. The producer-side node volume is taken from the selected SWEP B25T heat exchanger, the consumer-side volume is taken from the Stelrad radiator water content, and the pipe volumes are computed from the internal diameter and pipe length. Heat loss coefficients are derived from simple insulation and convection assumptions for the producer, cylindrical thermal resistance relations for the pipes, and a radiator rated-output approximation for the consumer.

The resulting modelling approach is sufficiently simple for online predictive control while still being tied to physical equipment dimensions and thermal-loss assumptions. This is advantageous because it provides a clear link between engineering design choices and the final closed-loop dynamic model.

Table 2.1: State, manipulated-variable, and disturbance definitions used in the DHN model.

Symbol	Meaning	Unit	Role in model
T_1	Producer node temperature	°C	State
T_2	Supply pipe temperature	°C	State
T_3	Consumer node temperature	°C	State
T_4	Return pipe temperature	°C	State
q_{12}	Flow from node 1 to node 2	kg/s	Manipulated variable
q_{23}	Flow from node 2 to node 3	kg/s	Manipulated variable
q_{34}	Flow from node 3 to node 4	kg/s	Manipulated variable
q_{41}	Flow from node 4 to node 1	kg/s	Manipulated variable
P	Producer thermal power input	W	Manipulated variable
P_d	Consumer thermal demand	W	Measured disturbance
$T_{a,i}$	Ambient temperature around node i	°C	Environmental parameter

Table 2.2: Physical properties and node volumes used in the DHN model.

Parameter	Symbol	Value	Description
Specific heat capacity of water	c_p	4180 J/(kg K)	Constant value used for all nodes
Water density (nodes 1,2)	$\rho_{1,2}$	968.39 kg/m^3	Density at ≈ 85°C
Water density (nodes 3,4)	$\rho_{3,4}$	980.45 kg/m^3	Density at ≈ 65°C
Producer node volume	V_1	7.66×10^{-3} m^3	Volume of SWEP B25T secondary circuit
Pipe volumes	$V_2 = V_4$	3.39×10^{-4} m^3	3 m pipe, inner diameter 12 mm
Consumer node volume	V_3	6.60×10^{-3} m^3	Water content of Stelrad radiator
Ambient temperatures	T_a	$[10, 10, 20, 10]^\top$ °C	Node-wise surrounding temperature

The thermal mass matrix of the system is defined as

$$M = \mathrm{diag}(m_1, m_2, m_3, m_4), \tag{2.36}$$

where the nodal masses are computed from the water density and node volumes

$$m_i = \rho_i V_i. \tag{2.37}$$

Using the parameters from Table 2.2, the nodal masses become

$$m_1 = 968.39 \times 7.66 \times 10^{-3} = 7.42 \text{ kg} \tag{2.38}$$

$$m_2 = 968.39 \times 3.39 \times 10^{-4} = 0.33 \text{ kg} \tag{2.39}$$

$$m_3 = 980.45 \times 6.60 \times 10^{-3} = 6.47 \text{ kg} \tag{2.40}$$

$$m_4 = 980.45 \times 3.39 \times 10^{-4} = 0.33 \text{ kg} \tag{2.41}$$

The heat loss contribution appearing in the state equations is written using the coefficient

$$\kappa_i = \frac{UA_i}{c_p}, \tag{2.42}$$

where UA_i represents the overall heat-transfer coefficient between node i and the ambient environment.

Table 2.3: Heat-loss coefficients used in the DHN model.

Parameter	Symbol	Value	Description
Producer heat loss	UA_1	0.77 W/K	Insulated rectangular heat exchanger body
Producer coefficient	κ_1	1.84×10^{-4} kg/s	UA_1/c_p
Pipe heat loss (supply)	UA_2	0.479 W/K	Insulated cylindrical pipe
Pipe heat loss (return)	UA_4	0.479 W/K	Same pipe geometry
Pipe coefficients	$\kappa_2 = \kappa_4$	1.15×10^{-4} kg/s	UA/c_p
Radiator heat transfer	UA_3	34.64 W/K	Based on rated radiator output
Consumer coefficient	κ_3	8.29×10^{-3} kg/s	UA_3/c_p

The heat-loss parameters UA_i represent the effective thermal conductance between each node and its surrounding environment. These values are obtained using simplified thermal-resistance models based on the geometry of the corresponding components: the plate heat exchanger for the producer node, insulated cylindrical pipes for the transport sections, and the radiator heat output specification for the consumer node. The coefficients κ_i used in the temperature dynamics are obtained by dividing UA_i by the specific heat capacity of water.

Chapter 3

NMPC Formulation

3.1 Controller structure

The nonlinear controller is constructed using MATLAB's `nlmpc` object. The state dimension is $n_x = 4$, the output dimension is $n_y = 4$, the number of manipulated variables is $n_{mv} = 5$, and the number of measured disturbances is $n_{md} = 1$. The sample time is chosen as $T_s = 60$ s, the prediction horizon is $N_p = 10$, and the control horizon is $N_c = 5$.

These settings imply that the controller predicts the plant over a 10-minute future window and optimizes the first five future control moves explicitly. After the control horizon is exhausted (meaning optimization does not find solution in this window), then the remaining future inputs are assumed constant (or last solution) according to the standard piecewise-constant MPC parameterization.

Table 3.1: Key NMPC tuning settings used in the simulation.

Setting	Value	Interpretation
Sample time	60 s	Controller update interval
Prediction horizon	10 steps	Future prediction length
Control horizon	5 steps	Number of optimized control moves
States	4	T_1, T_2, T_3, T_4
Outputs	4	All states are output variables
Manipulated variables	5	$q_{12}, q_{23}, q_{34}, q_{41}, P$
Measured disturbances	1	P_d
Solver	`fmincon` with SQP	Nonlinear constrained optimization
Maximum iterations	50	Per sampling instant
Optimality tolerance	10^{-3}	Stopping criterion
Step tolerance	10^{-6}	Step-size stopping criterion
Suboptimal solution	Enabled	Improves robustness of closed-loop execution

3.2 State and output functions

The controller is linked to the user-defined state and output callbacks

$$\dot{x}(t) = f\left(x(t), \tilde{u}(t)\right), \qquad y(t) = h\left(x(t), \tilde{u}(t)\right) = x(t). \tag{3.1}$$

In the MATLAB implementation, the function `DHN_mpcStateFcn` reshapes the state and input vectors, extracts q, P, and P_d, and then calls the plant model `DHN_ct_plant`. The output function `DHN_mpcOutputFcn` simply returns the state vector itself.

3.3 Objective function

The standard MPC cost is replaced by a user-defined custom function. The implemented objective is

$$J = \sum_{k=0}^{N_p-1} \left(P_k - P_{d,k}\right)^2. \tag{3.2}$$

This cost is intentionally simple: it penalizes the mismatch between producer heat input and consumer heat demand over the prediction horizon. Since the cost does not include an explicit temperature-tracking term, a direct penalty on flow magnitude, or a penalty on control-rate changes, the optimizer is free to choose any admissible flow pattern and temperature evolution as long as the power mismatch is minimized while satisfying the constraints.

3.4 Equality constraints

The custom equality constraint function enforces

$$q_{12} = q_{23} = q_{34} = q_{41}. \tag{3.3}$$

In implementation form, the equality constraint vector is

$$c_{eq} = \begin{bmatrix} q_{12} - q_{23} \\ q_{23} - q_{34} \\ q_{34} - q_{41} \end{bmatrix} = 0. \tag{3.4}$$

This is a physically meaningful simplification for a single-loop DHN with one common circulation stream. By enforcing identical flow rates in all four branches, the controller is prevented from choosing inconsistent mass transport levels in different segments of the same loop. At the same time, this equality constraint reduces the number of truly independent flow decisions and makes the result easier to interpret: all flow trajectories should collapse onto the same profile if the optimization is working correctly.

3.5 Inequality constraints

The custom inequality constraints are formulated in the standard form

$$c(\cdot) \leq 0. \tag{3.5}$$

Two conditions are enforced.

First, every flow must remain strictly positive above a small lower bound:

$$q_{ij} \geq q_{\varepsilon}, \qquad q_{\varepsilon} = 10^{-3} \text{ kg/s}, \tag{3.6}$$

which is implemented as

$$q_{\varepsilon} - q_{ij} \leq 0. \tag{3.7}$$

This avoids zero or negative circulation and guarantees a consistent transport direction around the ring.

Second, the producer power is constrained to remain at least as large as the heat demand:

$$P \geq P_d, \tag{3.8}$$

which is implemented as

$$P_d - P \leq 0. \tag{3.9}$$

Because the cost already minimizes $(P - P_d)^2$, the inequality in Equation (3.8) drives the optimal solution toward the boundary $P = P_d$ whenever no other competing objective exists. This explains the very close match observed later between the producer power and the disturbance profile.

3.6 Complete NMPC problem statement

Putting together the model, objective, and constraints, the implemented optimal control problem at each sampling instant can be written as

$$\min_{\{u_k\}_{k=0}^{N_c-1}} \quad \sum_{k=0}^{N_p-1} (P_k - P_{d,k})^2 \tag{3.10}$$

$$\text{subject to} \quad x_{k+1} = F(x_k, u_k, P_{d,k}), \tag{3.11}$$

$$q_{12,k} - q_{23,k} = 0, \tag{3.12}$$

$$q_{23,k} - q_{34,k} = 0, \tag{3.13}$$

$$q_{34,k} - q_{41,k} = 0, \tag{3.14}$$

$$q_\varepsilon - q_{12,k} \leq 0, \tag{3.15}$$

$$q_\varepsilon - q_{23,k} \leq 0, \tag{3.16}$$

$$q_\varepsilon - q_{34,k} \leq 0, \tag{3.17}$$

$$q_\varepsilon - q_{41,k} \leq 0, \tag{3.18}$$

$$P_{d,k} - P_k \leq 0, \tag{3.19}$$

where $F(\cdot)$ denotes the internally discretized version of the continuous-time nonlinear plant.

In the implemented controller, the plant model is specified in continuous time within the MATLAB script `Main_DHN_NMPC.m`. This is done using the command

```
nlobj.Model.IsContinuousTime = true;
```

When this option is enabled, the nonlinear model predictive control (NMPC) toolbox automatically performs the required time discretization internally during the prediction stage. Specifically, the continuous-time plant model provided through the user-defined state function is numerically integrated over each sampling interval using MATLAB's built-in integration routines. As a result, the discrete-time mapping $F(\cdot)$ appearing in

the prediction model

$$x_{k+1} = F(x_k, u_k, P_{d,k})$$

is obtained implicitly by integrating the continuous-time dynamics over one sampling period. This allows the controller to be formulated using a continuous-time physical model while still solving the optimal control problem in discrete time at each sampling instant.

3.7 Discussion of the chosen formulation

It is essential to recognize that the implemented NMPC does not yet solve a complete temperature-tracking control problem. Even though a temperature reference vector is supplied to the Simulink NMPC block, the standard cost has been replaced by a custom cost that depends only on $P - P_d$. As a result, the optimization objective focuses on matching heat production to demand rather than regulating T_1 through T_4 to desired setpoints.

Chapter 4

Simulation Setup

4.1 Closed-loop implementation in MATLAB/Simulink

The overall simulation is implemented in MATLAB/Simulink with the nonlinear plant represented by a MATLAB Function block and a state integrator. The `nlmpc` object is constructed in a script `Main_DHN_NMPC.m`, exported to the base workspace, and then used by the Simulink Nonlinear MPC Controller block.

The high-level simulation sequence is as follows:

1. define controller tuning parameters, initial conditions, and reference values,
2. create and configure the `nlmpc` object,
3. assign the continuous-time state and output callback functions,
4. replace the standard cost and standard constraints with custom user-defined functions,
5. export the controller and initialization variables to the workspace,
6. open the Simulink model and run the closed-loop simulation over one day.

The simulation length visible from the plots corresponds to a full 24-hour period, i.e., 86400 s. This is appropriate because the disturbance profile is also defined over a 24-hour interval and wrapped periodically.

4.2 Measured disturbance profile

The heat demand profile is generated by the function `Pd_24h`. It has a constant base value of 40 kW and two Gaussian-shaped demand peaks superimposed on the baseline:

- a morning peak centered at 07:30 with an amplitude of 20 kW,
- an evening peak centered at 19:00 with an amplitude of 30 kW.

The disturbance is evaluated on a 60-second grid and supplied to the controller through zero-order hold interpolation. This is a practical and MPC-friendly choice because it aligns the disturbance sampling with the controller sample time.

4.3 Plant subsystem

The internal plant subsystem is shown in Figure 4.1. The subsystem takes the flow vector q, producer power P, and demand P_d as inputs. It also receives the current state vector T through feedback. The MATLAB function block computes the state derivative $\dot{T}$, and an integrator block reconstructs the temperature states. This is fully consistent with the continuous-time state-space implementation of the DHN model.

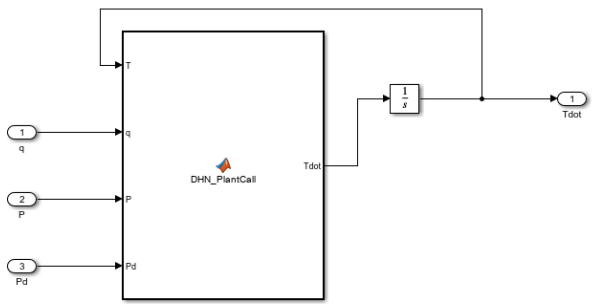

Figure 4.1: Plant subsystem used in Simulink. The MATLAB function block computes the temperature derivatives, which are then integrated to produce the state vector.

4.4 Initial conditions and references

The simulation starts from

$$T_1(0) = 85°\text{C},\ T_2(0) = 85°\text{C},\ T_3(0) = 65°\text{C},\ T_4(0) = 65°\text{C}. \tag{4.1}$$

The same values are also supplied as the initial output reference vector. This means that, from a modelling point of view, the intended thermal operating condition is a hot producer side and a cooler consumer/return side. However, as already emphasized in

Chapter 3, the custom cost function does not directly penalize deviations from these temperature references.

4.5 Solver configuration

The nonlinear optimization problem at each sample is solved using `fmincon` with the sequential quadratic programming (SQP) algorithm. The display output is turned off, the maximum number of iterations is limited to 50, and the controller is allowed to use a suboptimal solution if the optimization does not fully converge within the specified tolerances. This is a pragmatic choice in receding-horizon control because it prioritizes closed-loop continuity and numerical robustness.

4.6 Expectation before simulation

Based on the mathematical formulation, one can anticipate the following before running the simulation:

- the optimization status should indicate repeated successful solution of the online optimization problem,
- the producer power P should closely track the demand P_d because the objective directly minimizes their mismatch,
- all four flow rates should become identical because of the hard equality constraints,
- the temperature states may not remain near their initial reference values because those references do not appear in the custom objective.

The results in the next chapter are interpreted against exactly these expected tendencies.

Chapter 5

Results and Discussion

5.1 Overview of obtained closed-loop behavior

The simulation results show a clear and internally consistent pattern. The optimization status remains constant and acceptable throughout the 24-hour run. The producer heat input follows the demand profile very closely. The four mass flow trajectories are practically identical, which confirms that the equality constraints are being enforced correctly. Meanwhile, the temperature states exhibit a pronounced initial transient and then settle to lower values than the initial references, with only minor visible variations over the daily cycle.

This combination of outcomes is highly informative: it indicates that the controller is functioning correctly with respect to the problem it was asked to solve, but that the chosen optimization problem is narrower than a full temperature-regulation problem.

5.2 Optimization status

Figure 5.1 shows the optimization status signal reported by the Simulink NMPC block.

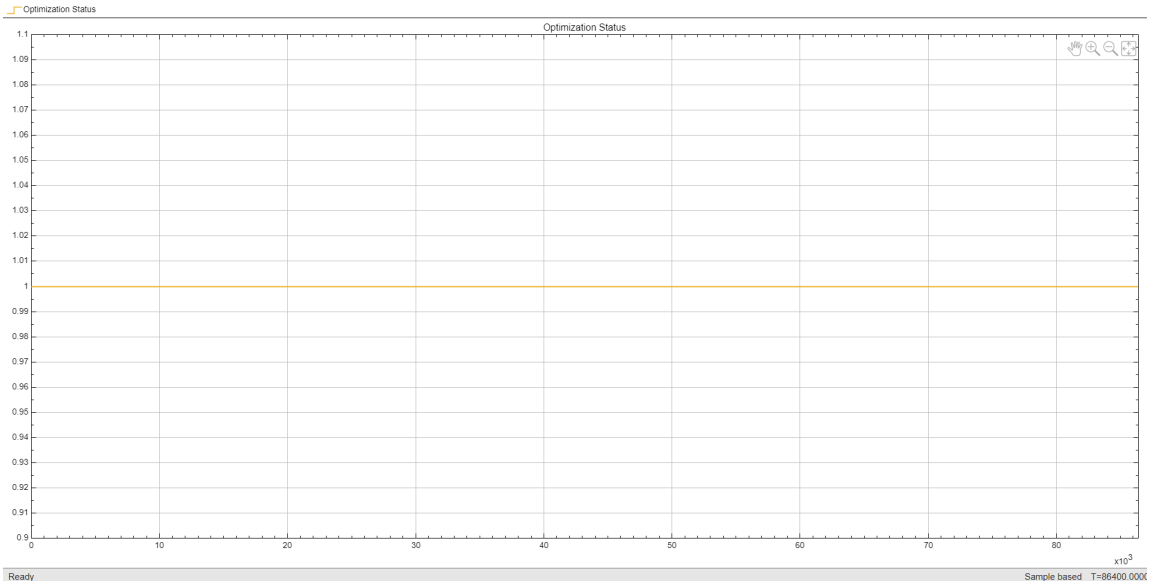

Figure 5.1: Optimization status during the full 24-hour simulation.

The status remains at a constant value of 1 throughout the simulation horizon. In the context of this study, this indicates that the controller returned an admissible optimization result at each sampling instant and that the online nonlinear program remained numerically well behaved during the entire run. This is an important result because it confirms that the model, custom callbacks, and solver settings are sufficiently consistent for closed-loop execution.

5.3 Flow-rate trajectories

The four flow-rate trajectories are shown in Figures 5.2–5.5. All four curves have the same shape, which is exactly what should happen when the equality constraints $q_{12} = q_{23} = q_{34} = q_{41}$ are active.

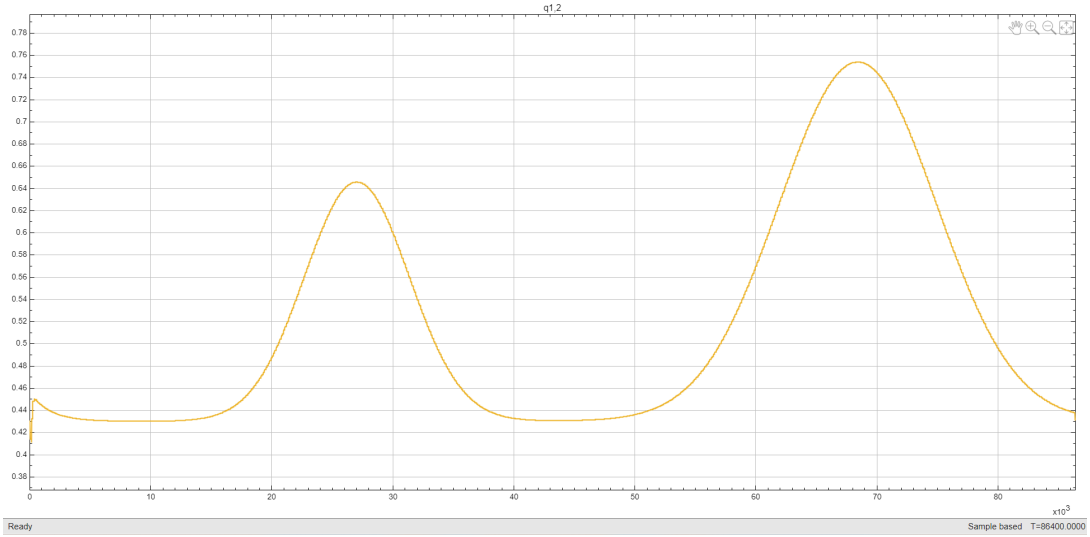

Figure 5.2: Manipulated flow rate q_{12} from node 1 to node 2.

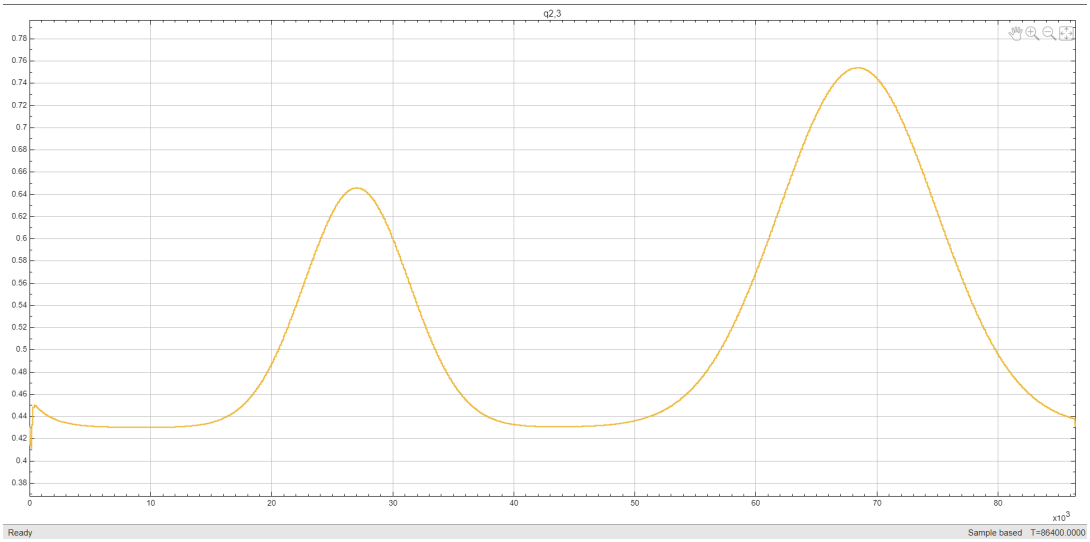

Figure 5.3: Manipulated flow rate q_{23} from node 2 to node 3.

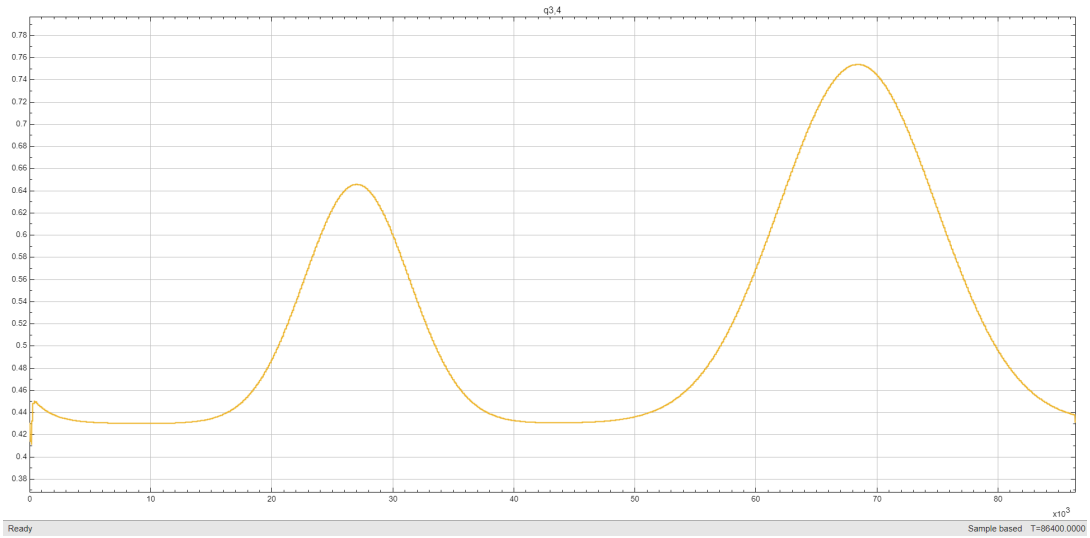

Figure 5.4: Manipulated flow rate q_{34} from node 3 to node 4.

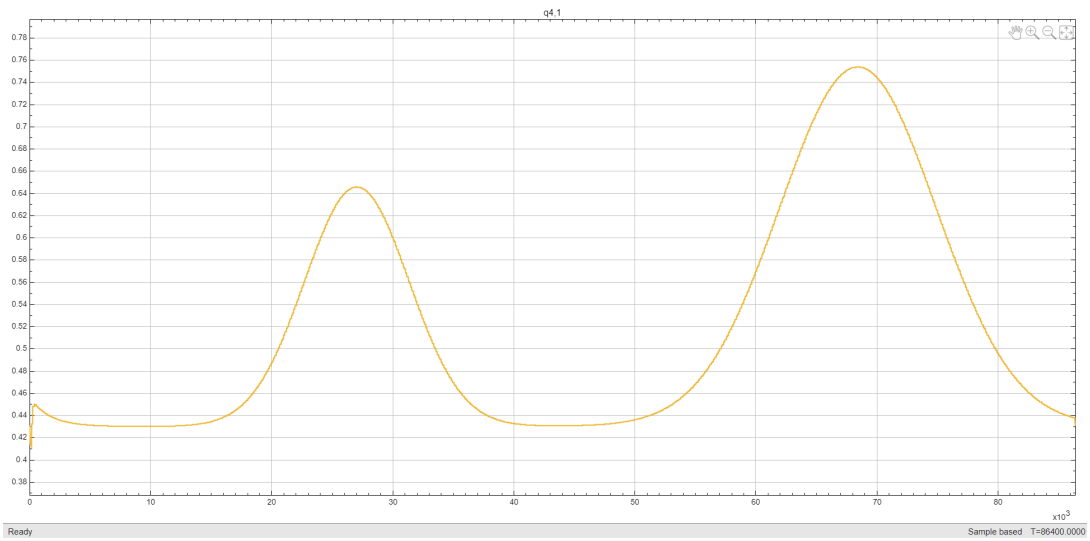

Figure 5.5: Manipulated flow rate q_{41} from node 4 to node 1.

The flow rates start near a moderate positive value and then increase during the morning and evening demand peaks. The evening peak is larger than the morning one, which mirrors the shape of the demand profile. Because the flow lower-bound constraint is active only as a minimum admissible circulation condition, the optimizer is free to choose higher values when needed. The result suggests that circulation is adjusted in a coordinated manner to support the heat-transport requirement under varying demand.

At the same time, it should be emphasized that the current cost function does not penalize large flow magnitudes. Therefore, the absolute flow level is not the outcome of a trade-off between pumping effort and thermal performance; it is simply whatever admissible value helps the optimizer satisfy the demand-oriented objective. In a more complete formulation, one would typically add a term such as $\sum q_k^2$ or $\sum \Delta q_k^2$ to avoid unnecessarily

large circulation rates.

5.4 Power and disturbance trajectories

The producer power input P and the measured demand disturbance P_d are shown in Figures 5.6 and 5.7.

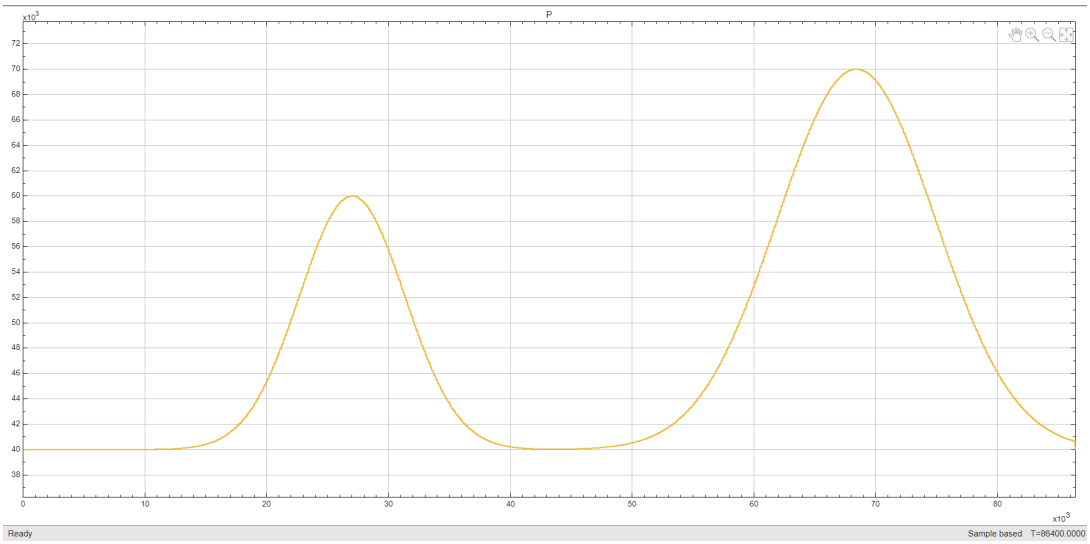

Figure 5.6: Producer heat input P over the 24-hour simulation.

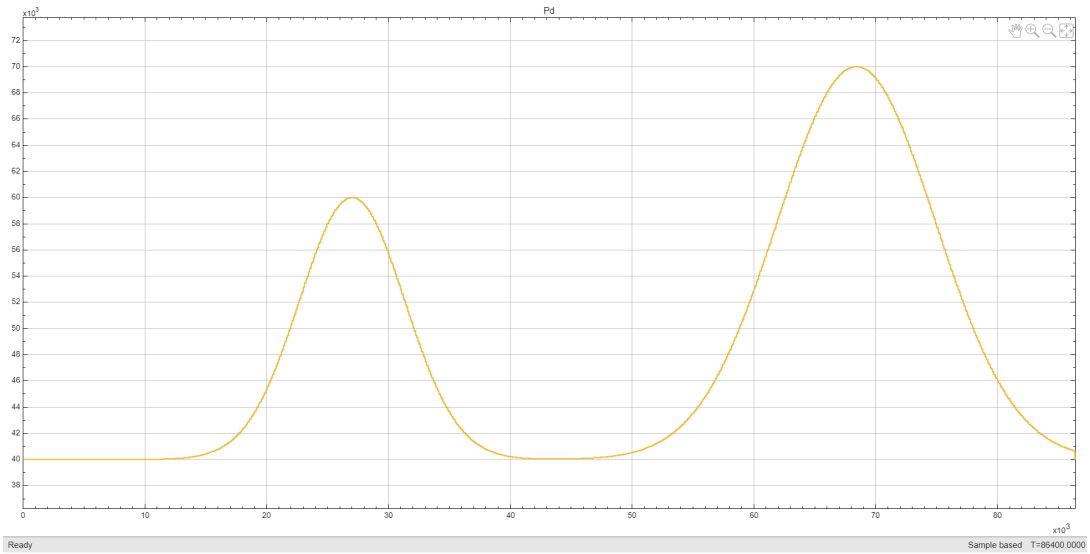

Figure 5.7: Measured disturbance profile P_d applied at the consumer node.

The two plots are visually almost identical. This is an expected consequence of the chosen optimization problem. The cost function minimizes the squared mismatch between P and P_d, while the inequality constraint enforces $P \geq P_d$. Together, these two ingredients make the optimal solution lie very close to the boundary $P = P_d$. In other words, the

optimizer is encouraged to supply exactly the amount of thermal power demanded by the load, without deliberate overproduction.

This result is good from a demand-following point of view. It shows that the controller can reconstruct the daily disturbance profile through the manipulated producer input. However, it also reveals a structural limitation: if the thermal states require additional energy beyond the instantaneous demand—for example, to maintain a high supply temperature or to preheat the network ahead of an expected future peak—the present objective does not explicitly reward such behavior. This is one of the key insights emerging from the simulation.

5.5 Temperature trajectories

The state trajectories are shown in Figures 5.8–5.11.

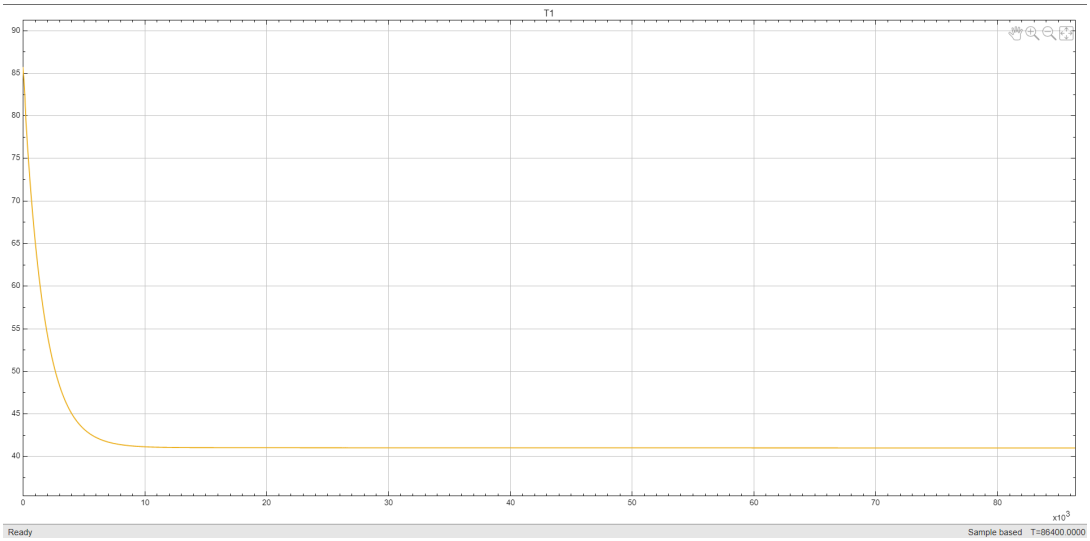

Figure 5.8: Temperature trajectory at node 1, T_1.

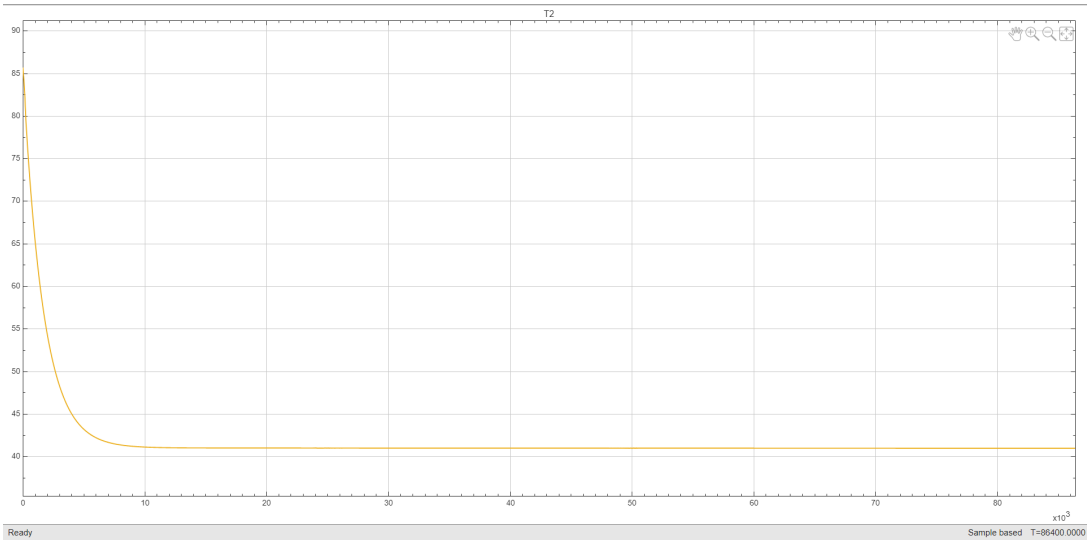

Figure 5.9: Temperature trajectory at node 2, T_2.

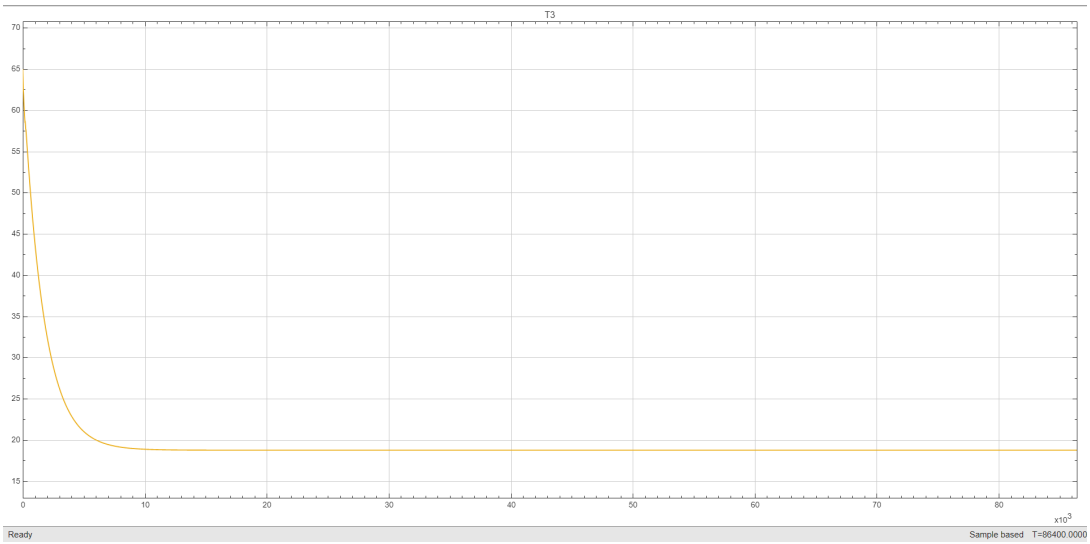

Figure 5.10: Temperature trajectory at node 3, T_3.

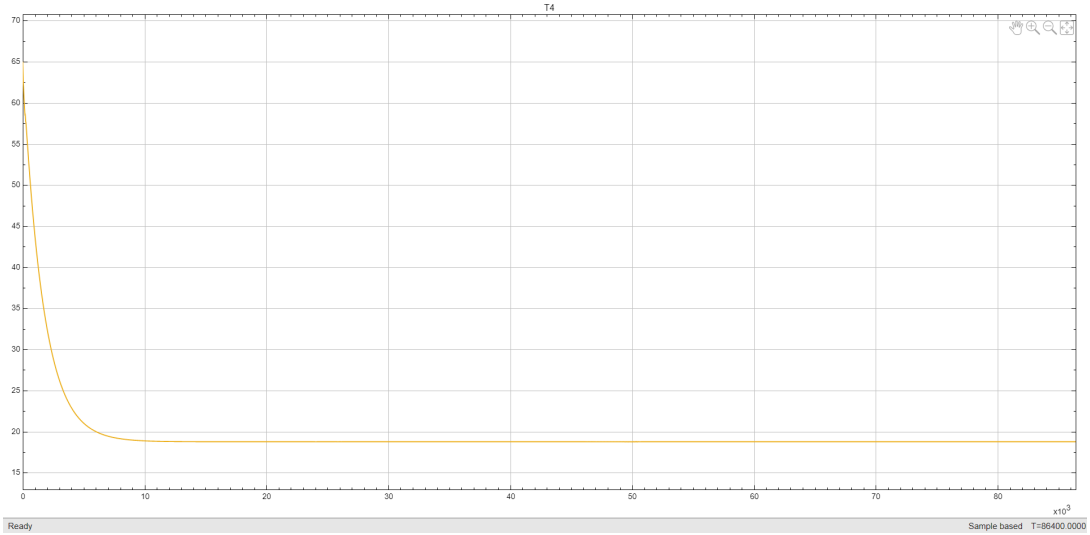

Figure 5.11: Temperature trajectory at node 4, T_4.

A strong initial transient is observed in all four states. The producer-side temperatures T_1 and T_2 fall rapidly from their initial value near 85°C to a much lower quasi-steady level around the low-forties. Likewise, the consumer-side and return-side temperatures T_3 and T_4 decrease from about 65°C to values around the high teens. After this transient, the curves appear almost flat on the plotted scale, meaning that any remaining daily variation is very small relative to the initial drop.

This behavior is the single most important result in the entire study because it shows that the current controller is not actually regulating the temperatures to the reference vector $[85, 85, 65, 65]^\top$. The reason is not numerical failure, modelling inconsistency, or optimization breakdown. Rather, it is a direct consequence of the custom cost design. Once the standard tracking cost is removed and replaced by the power-mismatch objective, the optimizer has no direct incentive to keep the states near the reference. As long as the demand is met and the constraints are satisfied, lower temperature operating points can become acceptable to the optimization problem. Or the chosen references are "wrong" in comparison to heat demand required by consumer as per optimization low temperature values can fulfill that demand.

From a control-engineering standpoint, this is an excellent demonstration of a core MPC principle: the controller optimizes exactly what the objective function asks for, not what the designer informally hopes for. Since temperature deviation is not penalized, temperature tracking is not guaranteed.

5.6 Discussion

The discussion can be structured around four main observations.

5.6.1 Observation 1: the NMPC implementation is numerically successful

The constant optimization status and smooth trajectories indicate that the online nonlinear optimization remains feasible and well behaved throughout the simulation. This validates the consistency of the custom MATLAB callbacks, the dimensions of the manipulated and disturbance inputs, and the closed-loop Simulink integration.

5.6.2 Observation 2: power tracking works exactly as designed

The close match between P and P_d is not accidental. It is the natural solution of an optimization problem whose only performance index is $(P - P_d)^2$ and whose inequality constraint enforces $P \geq P_d$. This confirms that the implemented cost and constraint logic are functioning properly.

5.6.3 Observation 3: equality constraints dominate the flow behavior

Because all four flow rates are forced to be equal, the optimizer effectively controls one common circulation level rather than four independent branch flows.

5.6.4 Observation 4: temperature references are not being controlled

This is the crucial insight. Although the Simulink NMPC block receives a reference input, the custom cost replaces the standard tracking cost. Therefore, the reference no longer influences the optimization unless it is reintroduced explicitly in the custom objective. The state trajectories clearly reflect this: the system settles to a thermal equilibrium driven by power balance, transport, and losses, rather than by the supplied reference vector. Or the chosen references are "wrong" in comparison to heat demand required by consumer as per optimization low temperature values can fulfill that demand.

5.6.5 Interpretation

This result can be turned into a strong analytical point rather than being treated as a problem. The present simulation demonstrates a demand-oriented NMPC formulation. It is successful in matching generation to consumption, but it is not yet a temperature-tracking NMPC. This distinction gives a natural pathway for future work:

- add explicit temperature tracking penalties such as $(T_k - T_k^{ref})^\top Q(T_k - T_k^{ref})$,
- add control effort or pumping penalties on the flow rates,
- add rate-of-change penalties to smooth actuator movement,
- include upper and lower bounds on temperature and power,
- include upper bounds on the flow rates,
- investigate terminal penalties or terminal constraints,
- extend the model to include pump dynamics, hydraulic losses, or transport delay.

In other words, the current report can be positioned as a first validated closed-loop framework that proves the computational feasibility of the NMPC architecture. The next stage would be to enrich the objective so that thermal regulation and operational efficiency are optimized simultaneously.

Chapter 6

Conclusion

This report presents a four-node district heating network controlled by nonlinear model predictive control in MATLAB/Simulink. The study began with a compact but physically meaningful thermal model based on node-wise energy balances. The corresponding continuous-time equations were derived explicitly and then assembled into a nonlinear state-space form suitable for the `nlmpc` framework.

A custom NMPC problem was then formulated. The manipulated variables consisted of four loop flow rates and one producer heat input, while the consumer heat demand was treated as a measured disturbance. The optimization objective penalized only the mismatch between produced and demanded thermal power. Equality constraints enforced a common circulation flow throughout the loop, and inequality constraints guaranteed positive flow and producer power not lower than demand.

The closed-loop simulation results showed that the optimization problem remained well posed and numerically solvable over the full 24-hour horizon. The producer power tracked the demand profile closely, and the four flow rates evolved identically as imposed by the equality constraints. These outcomes confirm that the controller implementation, plant model, and custom optimization callbacks are internally consistent.

At the same time, the temperature trajectories revealed that the node temperatures are not regulated to their nominal references. Instead, the temperatures undergo an initial transient and settle at lower quasi-steady values. Or because the chosen references are "wrong" in comparison to heat demand required by consumer as per optimization low temperature values can fulfill that demand. This behavior is fully consistent with the chosen objective function and is therefore a meaningful control-design insight rather than a simple deficiency. The present NMPC formulation should thus be interpreted as a demand-following controller rather than a full temperature-tracking controller.

The main conclusion is that the developed framework is a strong foundation for future research. It already demonstrates successful closed-loop nonlinear optimization with a

physically interpretable DHN model. Future improvements should enrich the custom cost and constraint structure so that demand matching, temperature tracking, pumping effort, actuator smoothness, and operational limits can all be addressed in a unified predictive-control design.

Bibliography

[1] SWEP International AB, *B25T Brazed Plate Heat Exchanger Datasheet*. Available online: https://www.hosbv.com/data/specifications/300535_Swep.pdf

[2] Stelrad Radiators, *Stelrad Classic Compact Radiator Technical Specifications*. Available online: https://www.stelrad.com/wp-content/uploads/2022/04/28968_Stelrad-HS_Classic_Web.pdf